# Student Enrollment & Course Management System (SECMS)

**A PROJECT REPORT**

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***in partial fulfilment for the award of the degree***

***of***

**BACHELOR OF SCIENCE**

**IN**

**COMPUTER SCIENCE**


**SSS COLLEGE OF ARTS, SCIENCE AND MANAGEMENT**

**(Affiliated to Thiruvalluvar University)**

**PG DEPARTMENT OF COMPUTER SCIENCE**

**User Story:**

The Student Enrollment & Course Management System is a comprehensive Salesforce application designed to streamline and automate the academic enrollment operations of an educational institute. This system enables administrators, enrollment officers, and instructors to efficiently manage student records, course offerings, enrollments, instructor assignments, approval processes, payments, and reporting—all within a unified Salesforce environment.

# 1. INTRODUCTION

## 1.1 Project Overview :

This project aims to centralize all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access, structured workflows, and accurate reporting. The system allows Enrollment Officers to manage student admissions, course registrations, and communication effectively, while instructors and administrators gain full transparency into course allocations and academic progress. The solution also enables automated instructor assignment, task creation, validation checks, and multi-level approval processes for enrollment management.

With Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization can ensure that all enrollment operations—from admission to course completion—are handled systematically and with continuous monitoring of key performance

indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

### 1.2 Purpose :

The primary purpose of SECMS is to replace manual and fragmented enrollment processes with a scalable, automated, and secure Salesforce solution that ensures:

Accurate student data management

Structured enrollment workflows

Automated approvals and validations

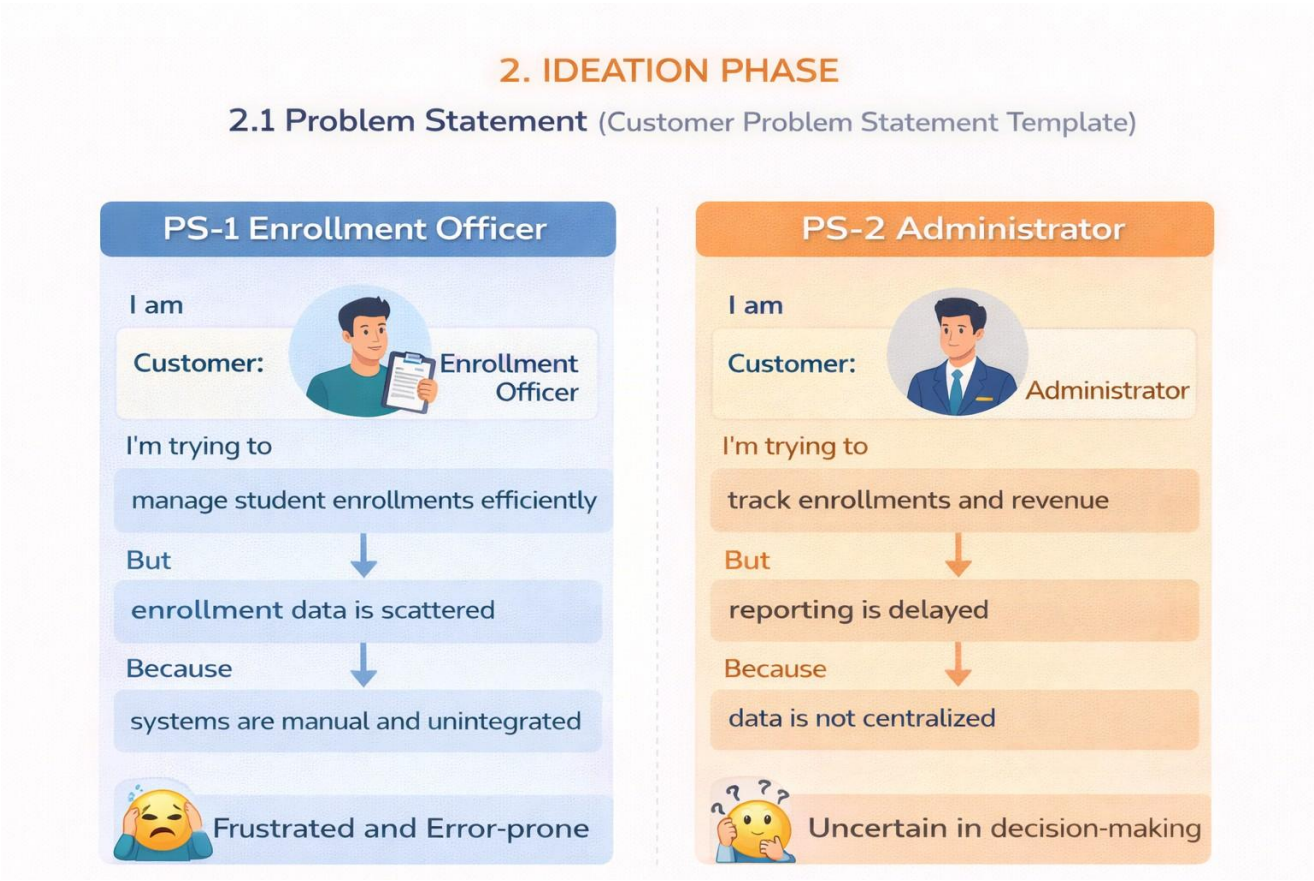
Transparent reporting and monitoring

## 2. IDEATION PHASE

### 2.1 Problem Statement (Customer Problem Statement Template)

| PS | I am (Customer) | I'm trying to | But | Because | Which makes me feel |
|---|---|---|---|---|---|

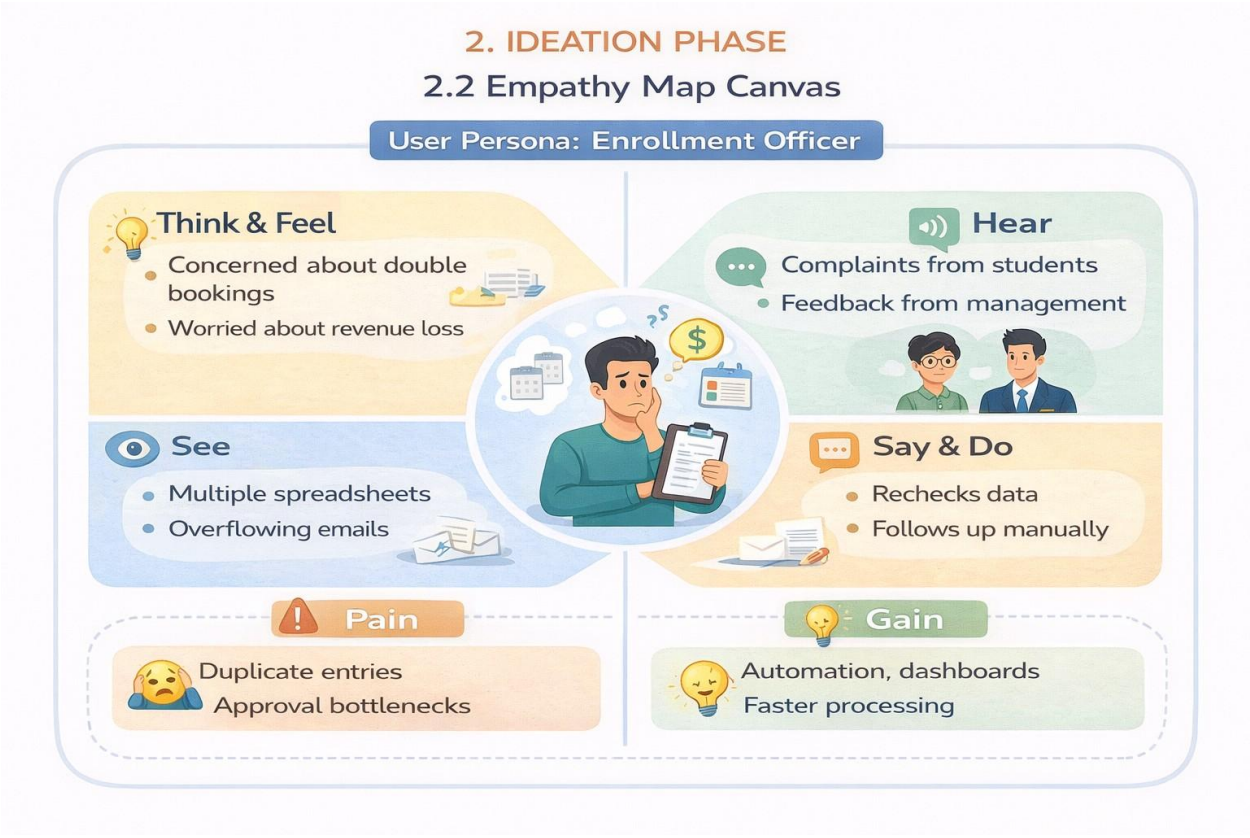
| | | | | | |
|---|---|---|---|---|---|
| PS-1 | Enrollment Officer | manage student enrollments efficiently | enrollment data is scattered | systems are manual and unintegrated | frustrated and error-prone |
| PS-2 | Administrator | track enrollments and revenue | reporting is delayed | data is not centralized | uncertain in decision-making |



## 2.2 Empathy Map Canvas

**User Persona:** Enrollment Officer

- **Think & Feel:** Concerned about double bookings and revenue loss
- **Hear:** Complaints from students and management
- **See:** Multiple spreadsheets and emails
- **Say & Do:** Rechecks data, follows up manually
- **Pain:** Duplicate entries, approval bottlenecks
- **Gain:** Automation, dashboards, faster processing



## 2.3 Brainstorming & Idea Prioritization

**Step 1 – Brainstormed Ideas**

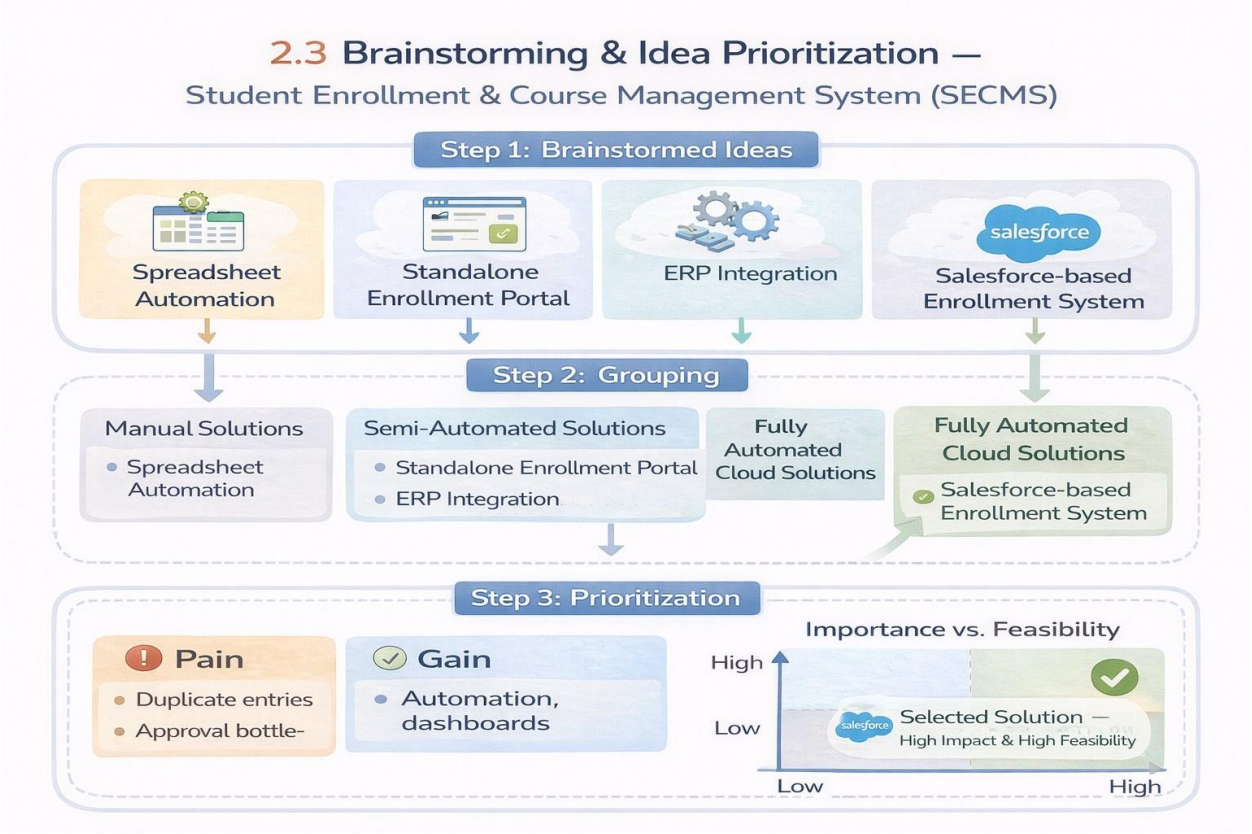
- Spreadsheet automation
- Standalone enrollment portal
- ERP integration
- Salesforce-based enrollment system

**Step 2 – Grouping**

- Manual solutions
- Semi-automated solutions
- Fully automated cloud solutions

**Step 3 – Prioritization**

Salesforce-based SECMS was selected due to **high feasibility** and **high impact**.

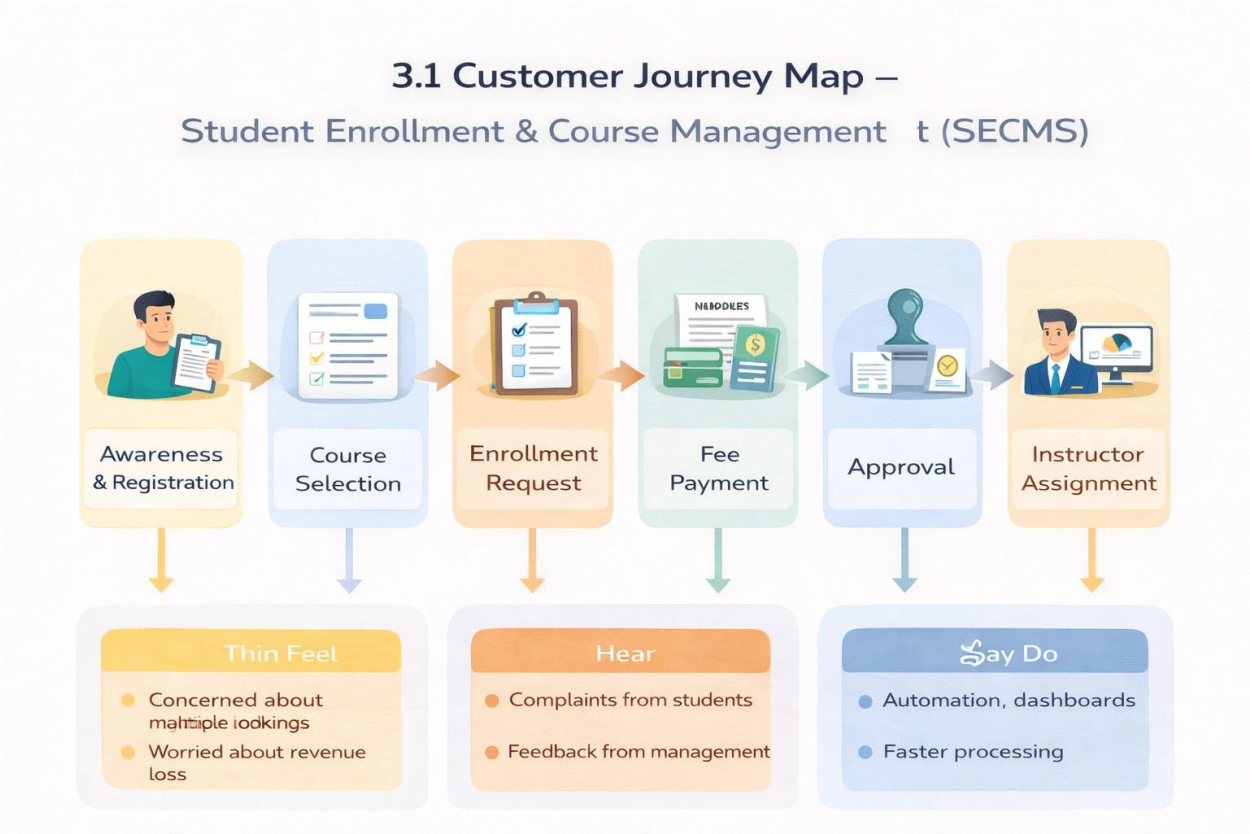



# 3.REQUIREMENT ANALYSIS PHASE

## 3.1 Customer Journey Map Stages

covered:

- Awareness & Registration
- Course Selection
- Enrollment Request
- Fee Payment
- Approval
- Instructor Assignment
- Reporting & Completion




## 3.2 Solution Requirements

**Functional Requirements**

| FR No | Epic | Description |
|---|---|---|
| FR-1 | Student Management | Manage student records |
| FR-2 | Course Management | Create and manage courses |
| FR-3 | Enrollment Processing | Request, approve, reject enrollments |
| FR-4 | Instructor Assignment | Auto-assign instructors |
| FR-5 | Reporting | Dashboards and reports |

## Non-Functional Requirements NFR-1: Usability

The system shall provide a simple, intuitive, and user-friendly interface that enables enrollment officers, administrators, instructors, and students to perform operations efficiently with minimal training. Navigation shall be organized through clearly labeled tabs, dashboards, and forms to reduce user effort and errors.

**Examples:**

- Quick student registration form
- One-click enrollment approval
- Role-based dashboards

## NFR-2: Security

The system shall ensure the confidentiality, integrity, and protection of student and institutional data through secure authentication, authorization, and access controls. Sensitive information such as personal data, enrollment history, and academic records shall only be accessible to authorized users.

**Examples:**

- Only admins can modify course details
- Instructors view only assigned students
- Students cannot access other student records

### NFR-3: Reliability

The system shall operate consistently and accurately without failures during enrollment processing, approvals, and reporting. Automation logic shall execute correctly to avoid duplicate or incorrect enrollments.

**Examples:**

- No duplicate course registration
- Accurate status updates after approvals
- Consistent report generation

### NFR-4: Performance

The system shall provide fast response times for all operations to ensure smooth and uninterrupted user experience, especially during peak admission periods.

**Examples:**

- Quick dashboard refresh
- Fast search for student records
- No lag during bulk uploads

### NFR-5: Scalability

The system shall support future growth in the number of students, courses, departments, and users without degradation in performance or functionality. It shall allow easy addition of new modules and features.

**Examples:**

- Add new departments without redesign
- Handle increased admissions annually

- Extend to mobile or portal integration

### NFR-6: Availability

The system shall be continuously accessible to users during institutional working hours and must minimize downtime. The platform shall support high availability for academic operations.

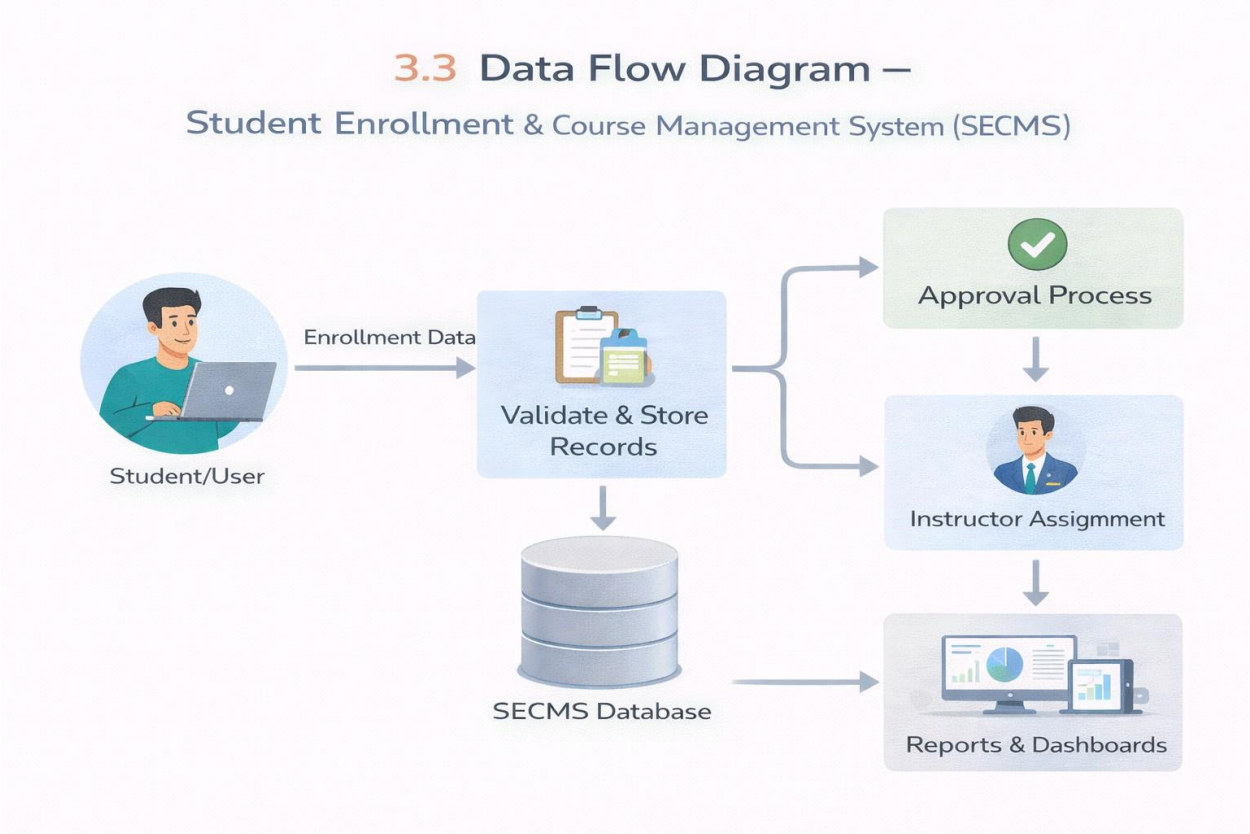
**Examples:**

- 24/7 cloud access
- Reliable during peak admission season
- Automatic backup of student records

## 3.3 Data Flow Diagram

- User inputs student and enrollment data
- System validates and stores records
- Approval process routes data
- Reports and dashboards consume processed data.

The system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.



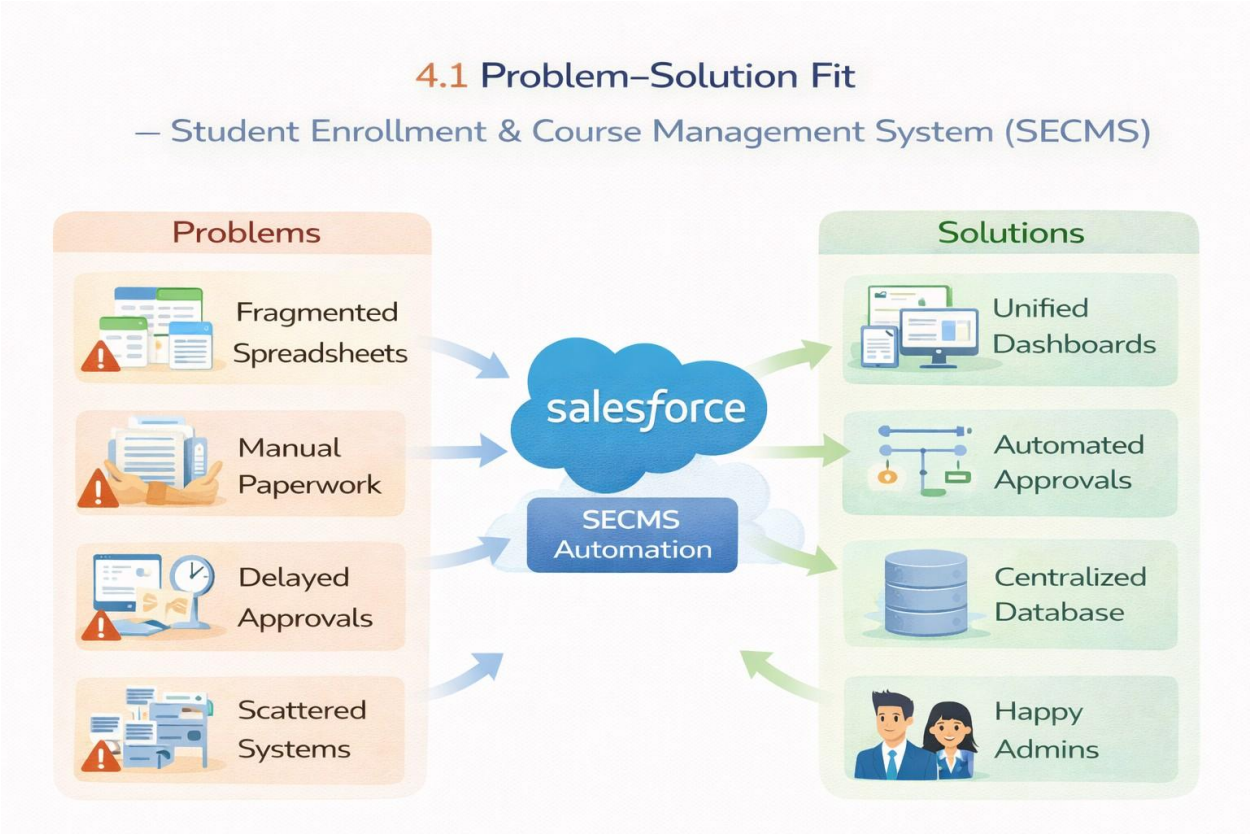

## 3.4 Technology Stack

| **Layer** | **Technology** |
|---|---|
| UI | Salesforce Lightning |
| Logic | Salesforce Flows |
| Database | Salesforce Objects |
| Automation | Validation Rules, Approval Process |
| Reporting | Reports & Dashboards |

# 4. PROJECT DESIGN PHASE

### 4.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.




## 4.2 Proposed Solution

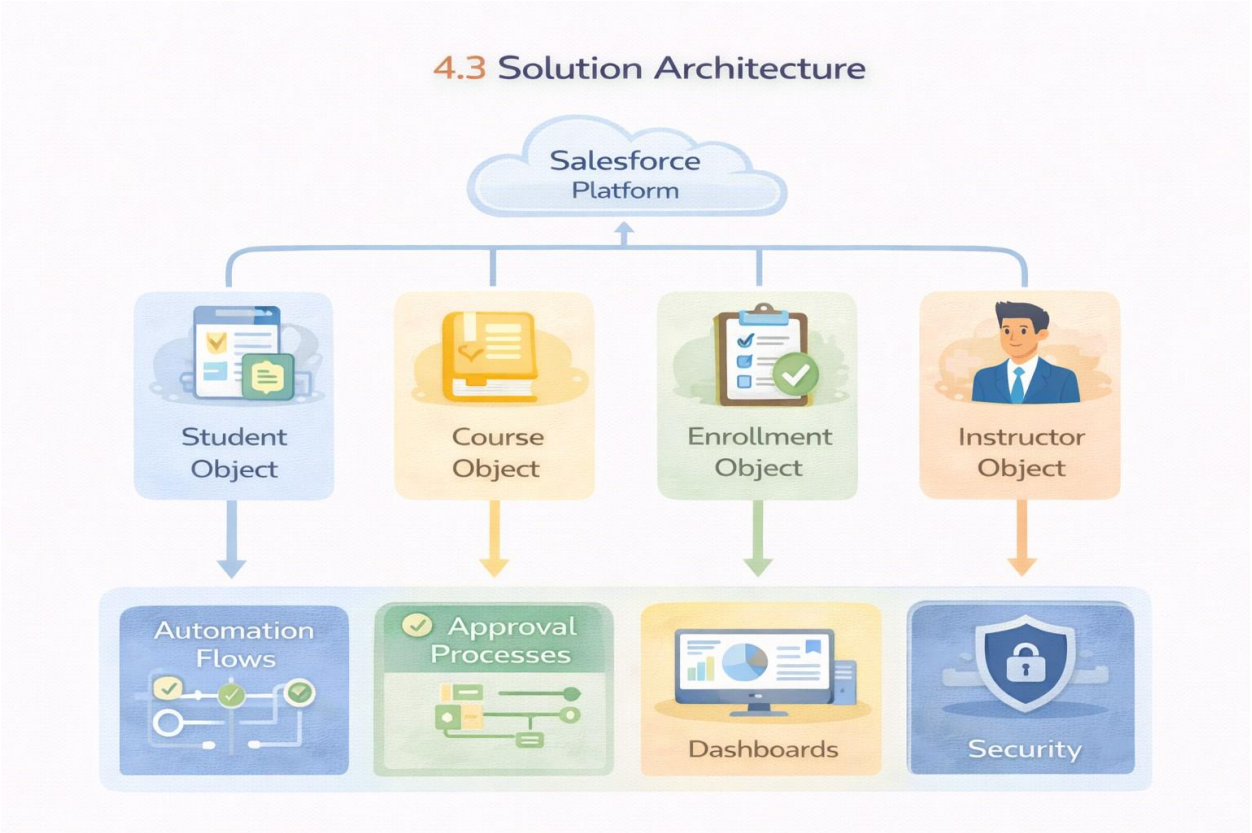
**Functional Requirements**

| **Parameter** | **Description** |
|---|---|
| Problem | Manual enrollment handling |
| Solution | Salesforce-based SECMS |

| | |
|---|---|
| Novelty | End-to-end automation |
| Social Impact | Improved student experience |
| Revenue Model | Institutional deployment |
| Scalability | Multi-department support |

## 4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers.



# 5. PROJECT PLANNING & SCHEDULING

## 5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

**Product Backlog, Sprint Schedule, and Estimation (4 Marks)**

Use the below template to create product backlog and sprint schedule

| Sprint | Functional Requirement (Epic) | User Story Number | User Story / Task | Story Points | Priority |
|---|---|---|---|---|---|
| Sprint-1 | Developer Setup | USN-1 | Course, Enrollment, and Instructor so that academic data can be stored and managed efficiently. | | |
| Sprint-2 | Data Modeling | USN-3 | As a user, I want dedicated tabs for each object so that I can easily navigate between students, courses, enrollments, and instructors. | 3 | High |
| Sprint-2 | Data Modeling | USN-4 | As an admin, I want to create relevant fields such as student details, course information, and enrollment status so that complete records can be captured accurately. | 5 | High |
| Sprint-2 | Data Modeling | USN-5 | As an enrollment officer, I want a centralized Lightning App so that I can access all academic operations from one interface. | 3 | High |
| Sprint-2 | Data Modeling | USN-6 | As an admin, I want to define relationships between students, courses, and enrollments so that the system can logically link and track academic activities. | 5 | Medium |
| Sprint-2 | Data Modeling | USN-7 | As a user, I want customized page layouts so that only relevant information is displayed, improving clarity and ease of data entry. | 5 | High |
| Sprint-3 | Automation | USN-8 | As a system, I want to validate input data such as mandatory fields and | 3 | High |

| | | | | | |
|---|---|---|---|---|---|
| | | | enrollment limits so that incorrect or incomplete records are prevented. | | |
| Sprint-3 | Automation | USN-9 | As a system, I want automated flows to handle enrollment processing and notifications so that manual effort is reduced and processes become faster. | 3 | High |
| Sprint-3 | Automation | USN-10 | As an administrator, I want an approval process for enrollment requests so that registrations are verified before confirmation. | 5 | High |
| Sprint-4 | Security | USN-11 | As an admin, I want to configure role-based security so that users can only access data relevant to their responsibilities. | 5 | High |
| Sprint-5 | Reports | USN-12 | As a manager, I want reports on student enrollments, course registrations, and instructor assignments so that I can monitor academic operations. | 4 | High |
| Sprint-6 | Dashboards | USN-13 | As a manager, I want visual dashboards showing enrollment | 4 | Medium |

**Project Tracker, Velocity & Burndown Chart: (4 Marks)**

## 6. Project Development Phase:

## Project Flow:

| Sprint | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) |
|---|---|---|---|---|---|
| Sprint-1 | 20 | 6 Days | 24 Oct 2022 | 29 Oct 2022 | 20 |
| Sprint-2 | 20 | 6 Days | 31 Oct 2022 | 05 Nov 2022 | |
| Sprint-3 | 20 | 6 Days | 07 Nov 2022 | 12 Nov 2022 | |
| Sprint-4 | 20 | 6 Days | 14 Nov 2022 | 19 Nov 2022 | |

**Milestone 1:** Salesforce Developer Account Creation

**Milestone 2:** Custom Object Creation (Student, Course, Enrollment, Instructor)

**Milestone 3:** Tabs Creation

**Milestone 4:** Lightning App Creation

**Milestone 5:** Fields Creation

**Milestone 6:** Create Relationships

**Milestone 7:** Page Layout Customization

**Milestone 8:** Validation Rules

**Milestone 9:** Flows (Automation)

**Milestone 10**: Approval Process

**Milestone 11:** Reports

**Milestone 12:** Dashboards

**Milestone 13**: Security Setup

**Milestone 14**: Conclusion

## What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Creating approval workflows
7. Developing reports & dashboards
8. Setting up security with profiles, permission sets, and sharing
9. End-to-end application testing

## Milestone 1-Salesforce :

## Introduction:

Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don’t know where you should start on your learning journey? If you’ve answered yes to any of these questions, then you’re in the right place. This module is for you.

Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features, that will help you sell smarter and faster. As you work toward your badge for this module, we'll take you through these features and answer the question, "What is Salesforce, anyway?".

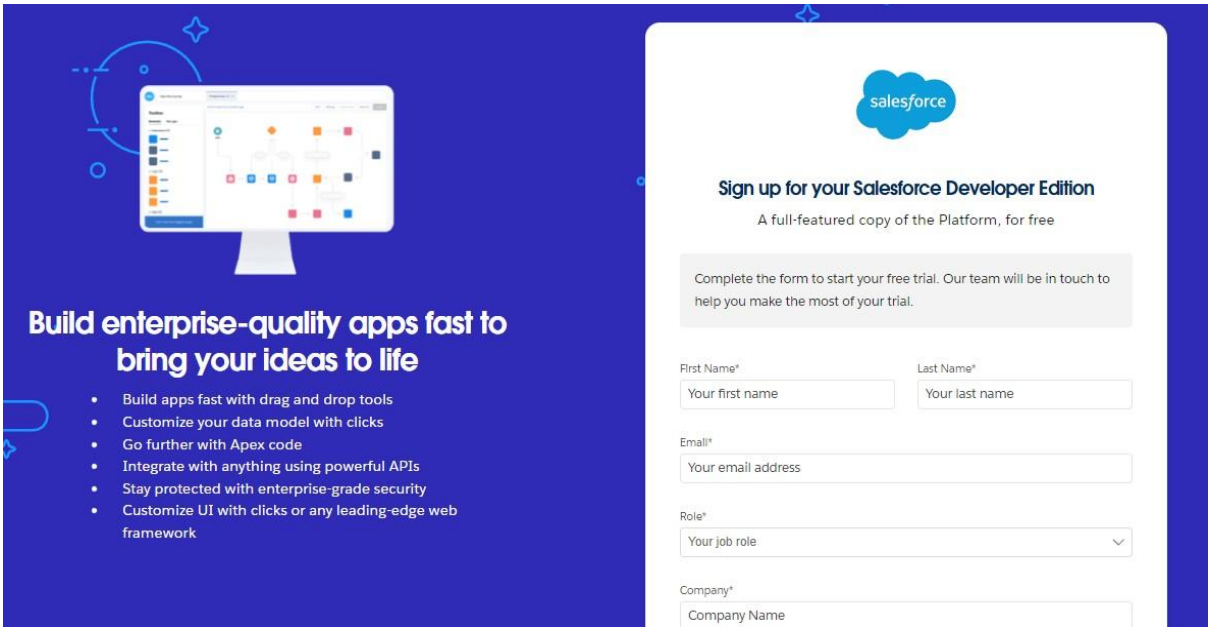
## What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud. So what does that really mean? Well, before Salesforce, your contacts, emails, follow-up tasks, and prospective deals might have been organized something like this:

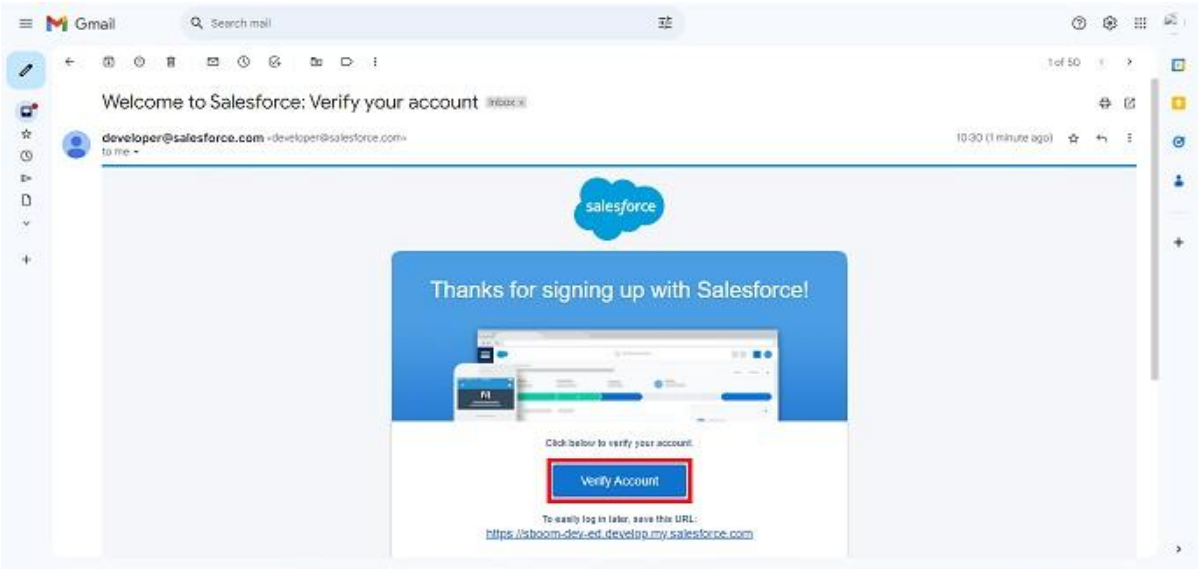
## Activity 1: Creating Developer Account:

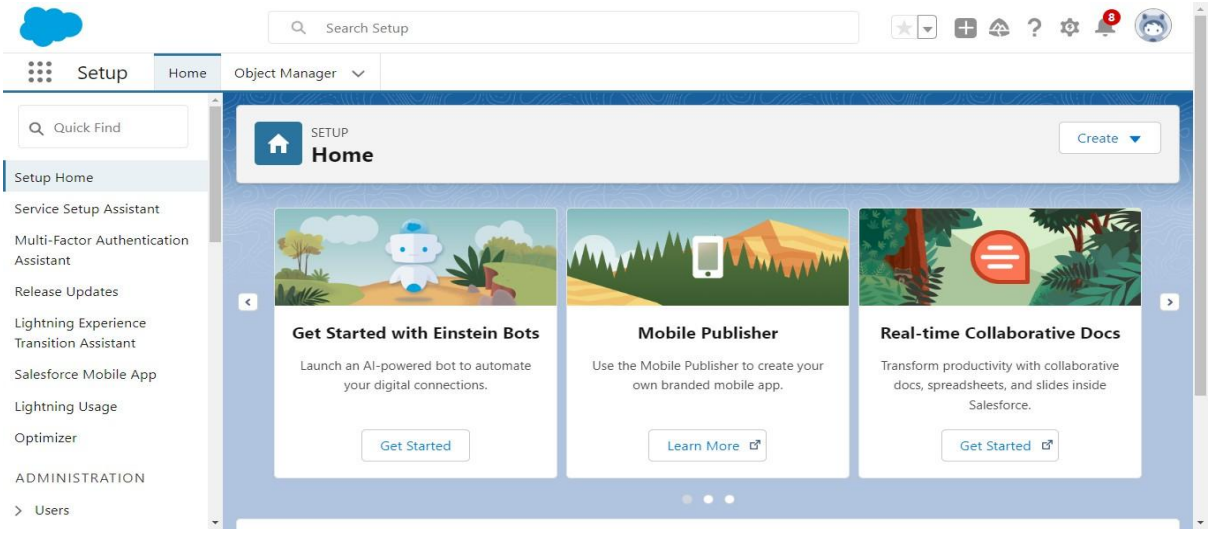
Creating a developer org in salesforce.




**Activity 2: Account Activation**:

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.

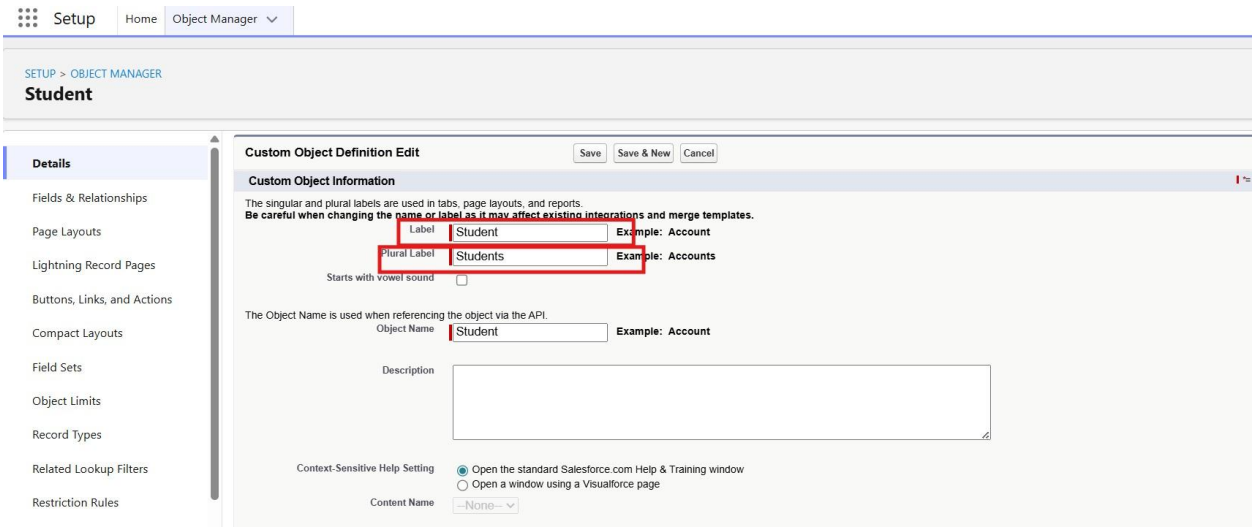





# Milestone 2 – Objects:

## Activity 1: Creating Student Object:

The purpose of creating the Student custom object is to store and manage information about students such as their personal details, contact information, and enrollment status.



## Milestone 3 - Fields:

**Table 1: Student Object Fields:**

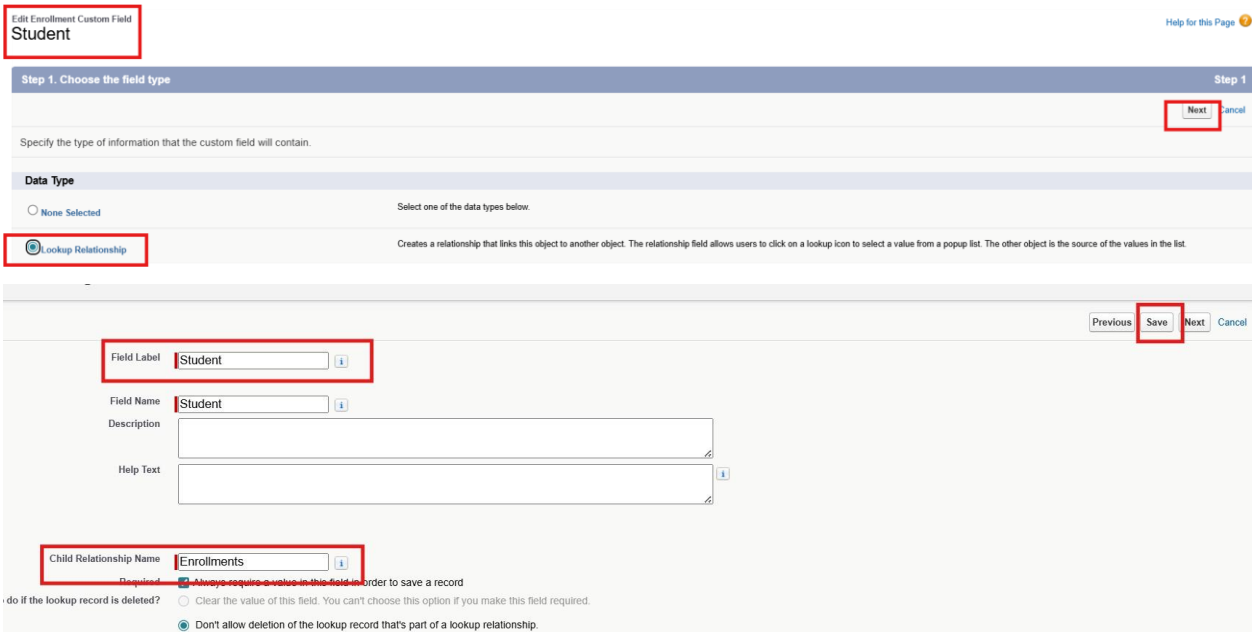
| Field Label | Data Type |
|---|---|
| Instructor Name | Text (Standard) |
| Instructor Code | Text |
| Expertise | Picklist |
| Phone | Phone |
| Email | Email |

**Table 4: Enrollment Object Fields:**

| **Field Label** | **Data Type** |
|---|---|
| Student | Lookup (Student) |

| | |
|---|---|
| Course | Lookup (Course) |
| Instructor | Lookup (Instructor) |
| Enrollment Date | Date |
| Enrollment Status | Picklist |
| Fees Paid | Checkbox |
| Total<br><br>Amount | Currency |
| Comments | Text Area |

## Milestone 4 – Create Relationships:




.

Edit Enrollment Custom Field
Course

Help for this Page

Step 1. Choose the field type

Step 1

Next Cancel

Specify the type of information that the custom field will contain.

Data Type

None Selected — Select one of the data types below.

Lookup Relationship — Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

Edit Relationship
Enrollment

Help for this Page

Step 2. Enter the label and name for the lookup field

Step 2 of 4

Previous Save Next Cancel

Field Label: Course

Field Name: Course

Description:

Help Text:

Child Relationship Name: Enrollments

Required: Always require a value in this field in order to save a record

What to do if the lookup record is deleted? Clear the value of this field. You can't choose this option if you make this field required.

Enrollment
New Relationship

Help for this Page

Step 2. Choose the related object — Step 2

Previous Next Cancel

Select the other object to which this object is related.

Related To: Instructor

Previous Next Cancel

Step 3. Enter the label and name for the lookup field — Step 3 of 6

Previous Next Cancel

Field Label: Instructor

Field Name: Instructor

Description:

Help Text:

Child Relationship Name: Enrollments

Required: Always require a value in this field in order to save a record

What to do if the lookup record is deleted?: Clear the value of this field. You can't choose this option if you make this field required.

Don't allow deletion of the lookup record that's part of a lookup relationship.

Auto add to custom report type: Add this field to existing custom report types that contain this entity

Enrollment
New Relationship

Help for this Page

Step 6. Add custom related lists — Step 6 of 6

Previous Save & New Save Cancel

Field Label: Instructor

Data Type: Lookup

Field Name: Instructor

Description:

Specify the title that the related list will have in all of the layouts associated with the parent.

Related List Label: Enrollments

Select the page layouts that should include this field. The field will be added as the last field in the first 2-column section of these page layouts. The field will not appear on any pages if you do not select a layout.

To change the location of this field on the page, you will need to customize the page layout. To change the values that appear, you will need to customize the Record Types.

| Add Related List | Page Layout Name |
|---|---|
| ☑ | Instructor Layout |

☑ Append related list to users' existing personal customizations

# Milestone 5 – Validation Rules:

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

**Student**

- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- **Validation Rules**

**Validation Rule Edit** — Save | Save & New | Cancel

Rule Name: Email_Must_Be_Student_Domain

Active: ☑

Description:

**Error Condition Formula**

Example: Discount_Percent__c>0.30 More Examples...

Display an error if Discount is more than 30%

If this formula expression is **true**, display the text defined in the Error Message area

Insert Field | Insert Operator

```
NOT(CONTAINS(Email__c, "@student.edu"))
```

Check Syntax

Functions: -- All Function Categories --

ABS, ACOS, ADDMONTHS, AND, ASCII, ASIN

Insert Selected Function

ABS(number)

Returns the absolute value of a number, a number without its sign

Help on this function

**Error Message**

**Example:** Discount percent cannot exceed 30%

This message will appear when Error Condition formula is **true**

Error Message: Email must be a valid student institution email

This error message can either appear at the top of the page or below a specific field on the page

Error Location: ○ Top of Page ◉ Field Email

Save | Save & New | Cancel

# Milestone 6 – Approval Process

**Step 2. Specify Entry Criteria** — Step 2

Previous | Save | Next

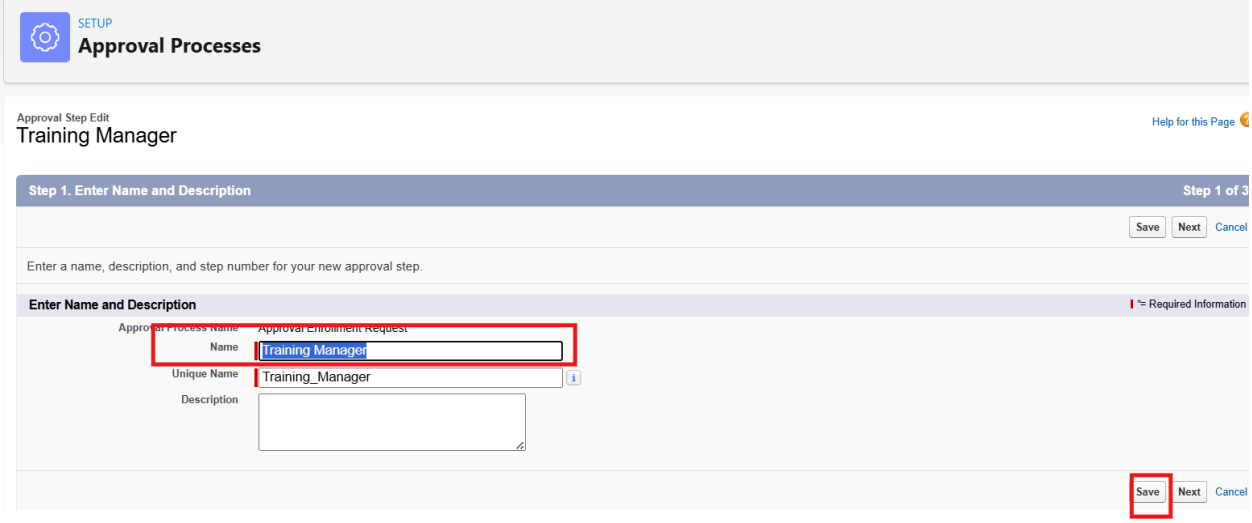
If only certain types of records should enter this approval process, enter that criteria below. For example, only expense reports from employees at headquarters should use this approval process.

**Specify Entry Criteria**

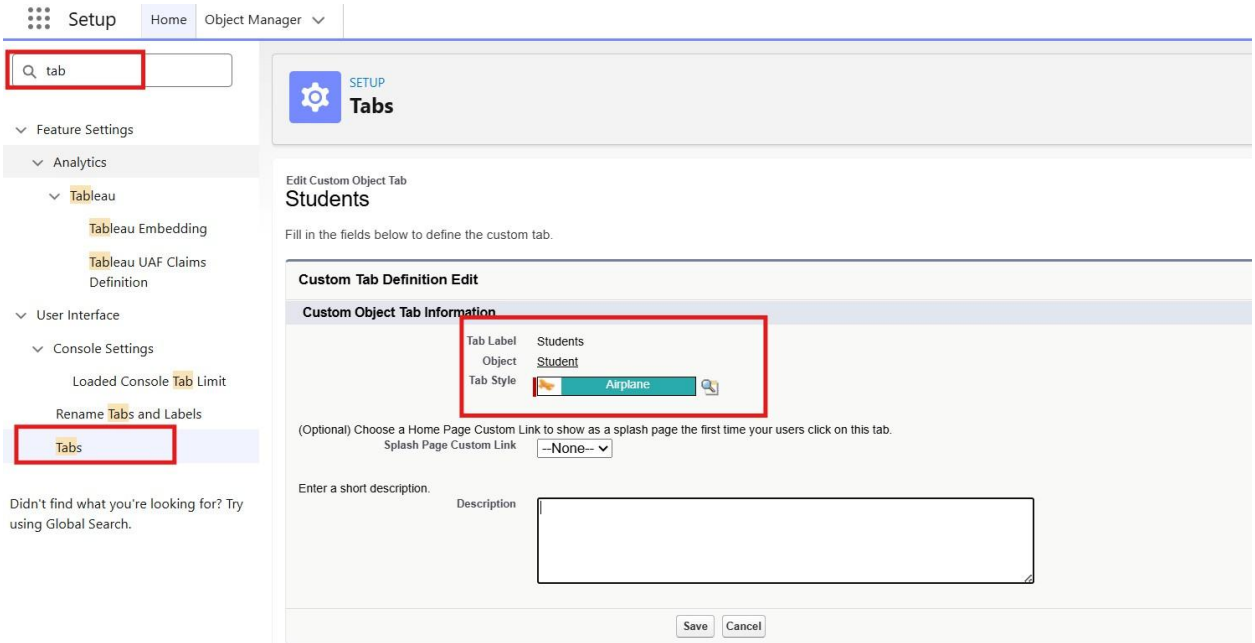
Use this approval process if the following criteria are met:

| Field | Operator | Value | |
|---|---|---|---|
| Enrollment: Enrollment Status | equals | Requested | AND |
| --None-- | --None-- | | AND |
| --None-- | --None-- | | AND |
| --None-- | --None-- | | AND |
| --None-- | --None-- | | |

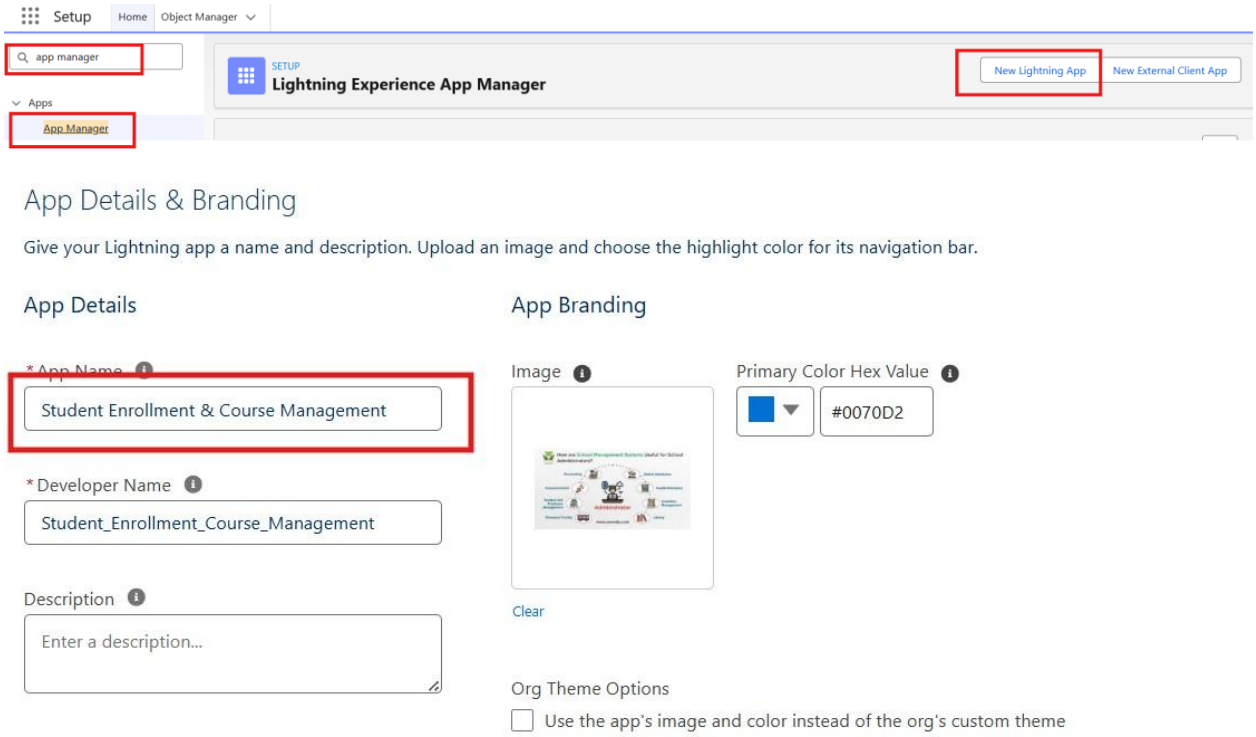
1. Add Approval Step:
2. Approver: specific Training Manager user.



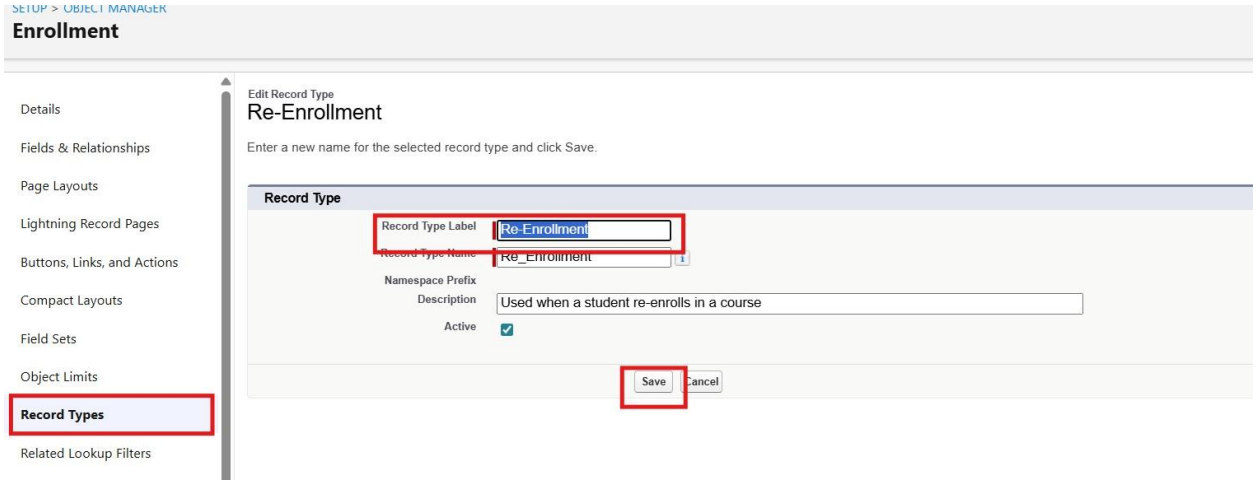

# Milestone 7- Tabs

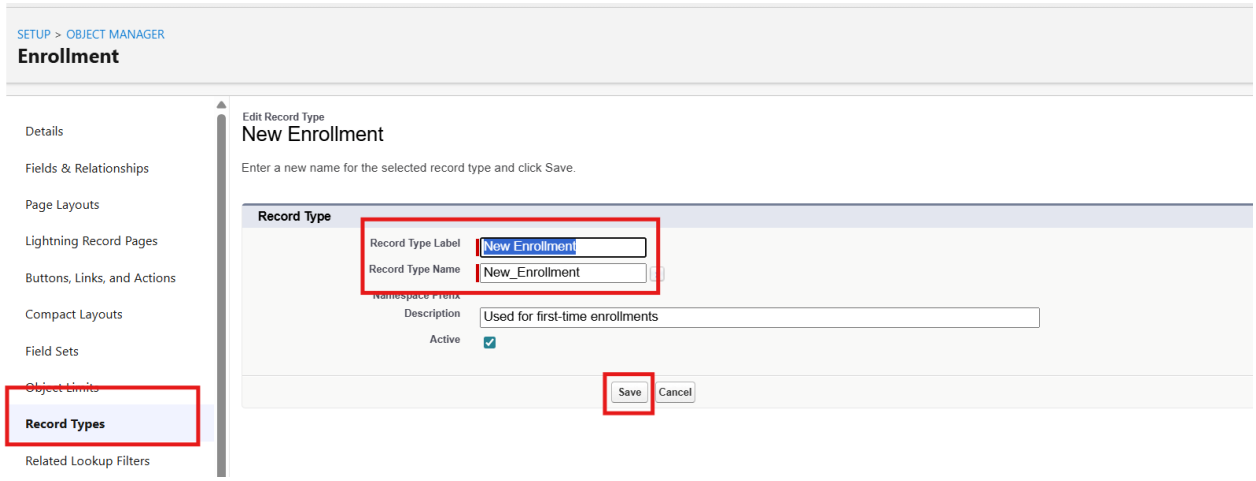



# Milestone 8 - The Lightning App

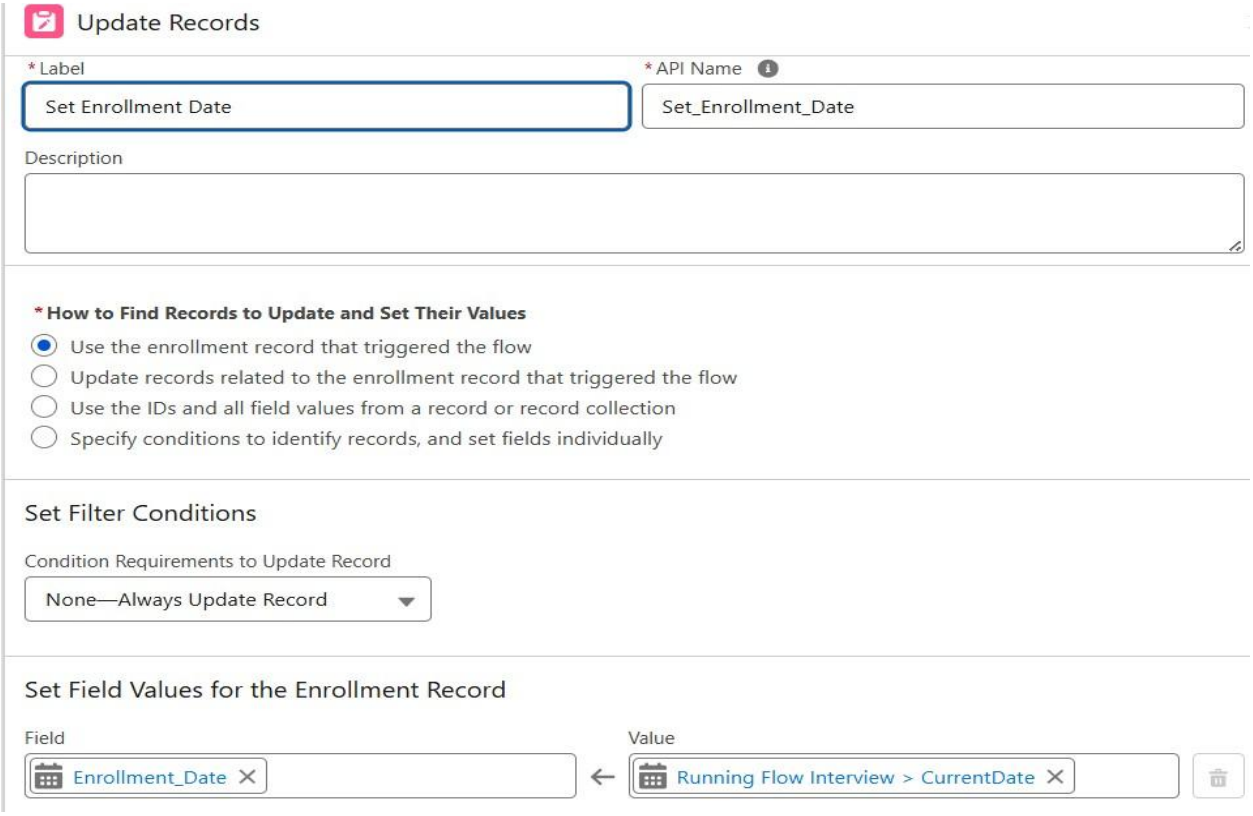


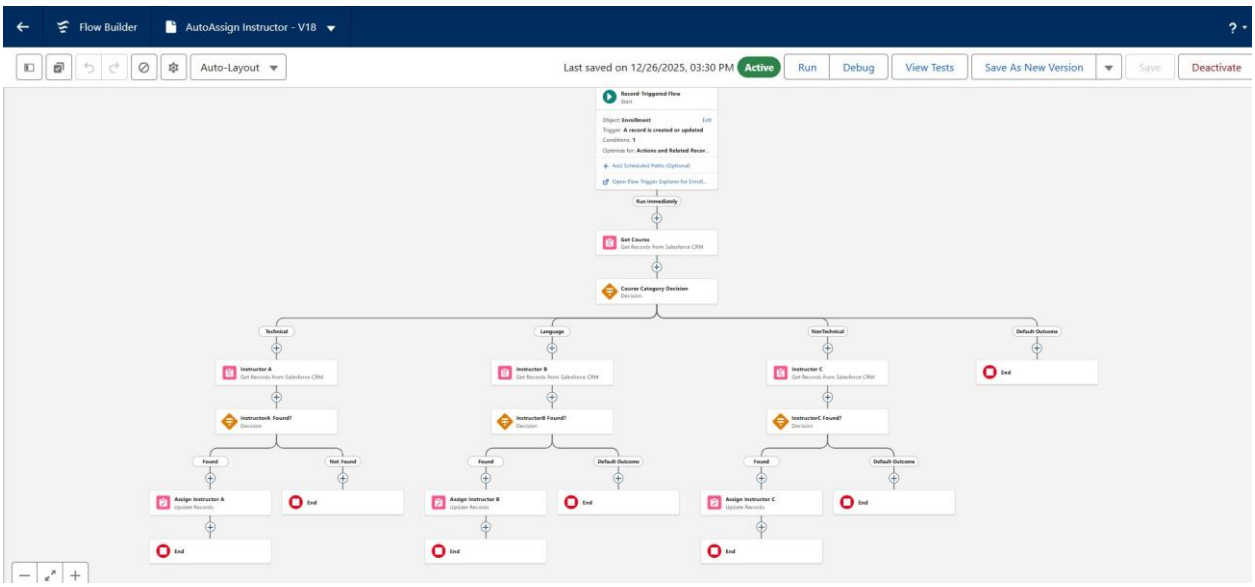

# Milestone 9 – Page Layouts and Record:

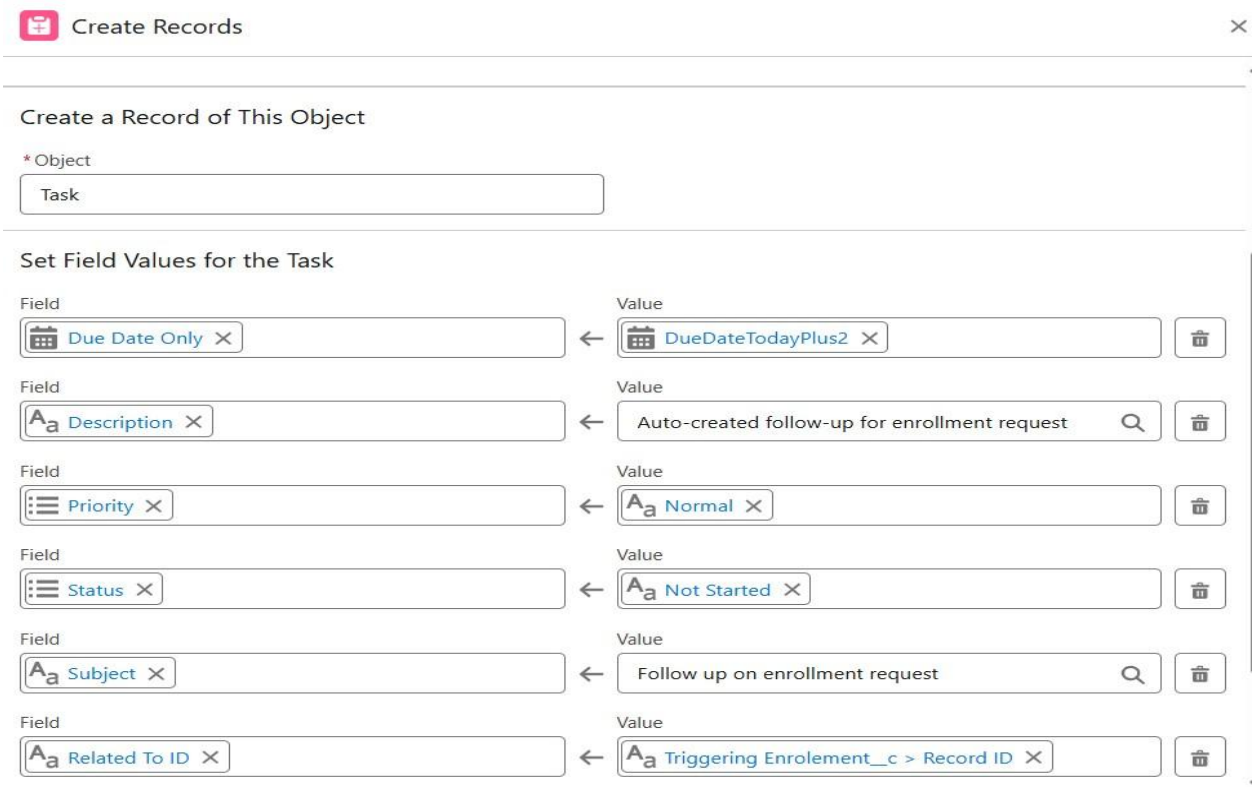


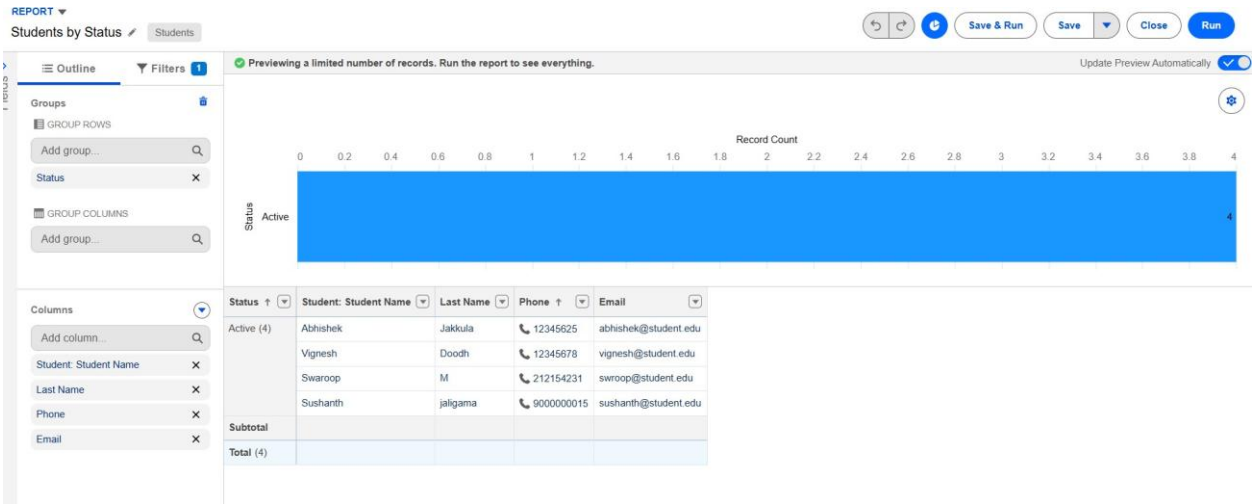
# Milestone 10 – Automation with Flow

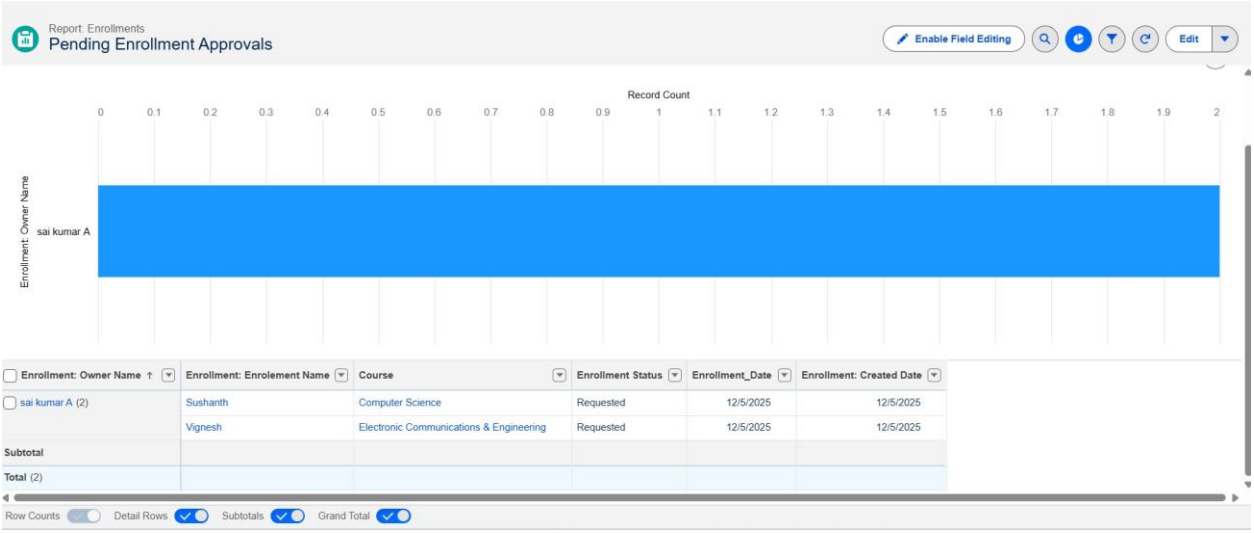


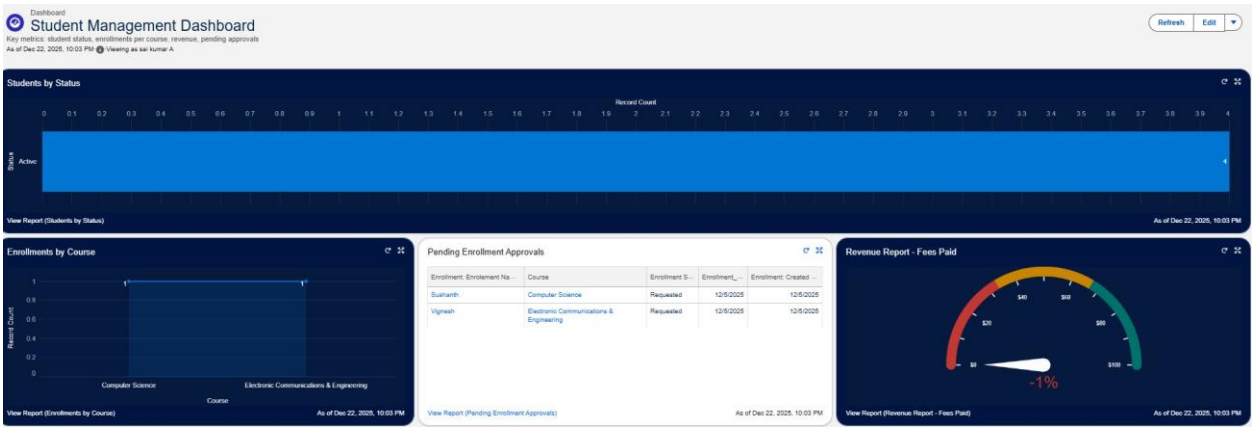



# Milestone 11 – Reports:






# Milestone 12 – Dashboards:



## 9. Conclusion:

The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle—from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, and reporting. By leveraging Salesforce's automation tools such as validation rules, flows, approval processes, and dashboards, the system improves data accuracy, operational efficiency, and decision-making. This project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

## 10. FUTURE SCOPE

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

## 11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

**Thank You**